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High-Precision Vehicle Localization Employing Li-Fi Supported V2V Communication and Sensor Fusion

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Abstract - Precise localization of vehicles are of outmost importance for realizing intelligent transportation systems. Traditional vehicle localization techniques based on Global Positioning System (GPS) technique and onboard sensors are often leads to inaccurate position estimation in case of dense urban areas, tunnels, and area with inadequate satellite coverage. To overcome these shortcomings, the present research addresses a cooperative vehicle-to-vehicle (V2V) localization method centered on Light Fidelity (Li-Fi) communication. In the suggested context, the adjacent vehicles exchange position and motion information through low-latency optical Li-Fi links, which ensures collaborative perception within the vehicular network. The shared data is integrated with locally obtained sensor measurements using a nonlinear sensor fusion technique to enhance localization accuracy and reliability. The simulation upshots revealed that the proposed cooperative Li-Fi-enabled localization approach shows improved positioning performance compared to conventional standalone localization methods. The study highlights the potential of Li-Fi-based V2V communication for enhancing localization reliability in connected and autonomous vehicle environments.

Index Terms - Vehicle Localization, Vehicle-to-Vehicle Communication, Li-Fi, Intelligent Transportation Systems

I. INTRODUCTION

The road transportation sector has been expanding rapidly as the number of vehicles increases and urban infrastructure continues to develop. Although this expansion enhances mobility and supports economic progress, it also creates serious challenges such as traffic congestion, road safety issues, and a higher risk of accidents. Global traffic safety studies indicate that many accidents occur because drivers react too late, lack timely information about nearby vehicles, or are unable to communicate effectively with other vehicles on the road [1]. In numerous situations, drivers may not notice sudden braking, obstacles, or dangerous road conditions ahead until it becomes difficult to respond safely. Consequently, the development of intelligent communication technologies that enable vehicles to exchange safety-related information in real time has become a critical focus area in modern Intelligent Transportation Systems (ITS) [2]. Vehicle-to-Vehicle (V2V) communication is a vital technology that allows vehicles to share important information such as speed, location, braking activity, and current traffic conditions. Through direct data exchange among nearby vehicles, V2V systems can alert drivers to possible dangers in advance and help minimize the likelihood of collisions [3]. Traditional V2V communication methods primarily depend on radiofrequency (RF) technologies, including Wi-Fi, dedicated short-range communication, and the emerging 5G communication networks. While these RF-based approaches enable wireless information transfer, they often encounter challenges in highly congested traffic environments. Issues such as spectrum overcrowding, electromagnetic interference, restricted bandwidth, and potential security risks can affect communication performance. As a result, the reliability of data transmission may decrease and latency may increase, which is particularly problematic for time-critical vehicular safety applications [4].

The aforesaid limitations can be addressed by an alternative wireless communication approach known as Light Fidelity (Li-Fi), which uses visible light to exchange information. Li-Fi technology works by rapidly modulating the light intensity produced by light emitting diodes (LEDs) or laser sources to encode digital data, which can then be detected by photodiodes or optical receivers [5]. Compared with conventional RF-based systems, Li-Fi provides several benefits, including very high data transfer speeds, reduced communication delay, resistance to electromagnetic interference, improved security, and more efficient use of the available spectrum. Because Li-Fi operates within the visible light spectrum, it does not interfere with existing RF communication networks and can function reliably in environments where radio signals may perform poorly [6-7].

II. LITERATURE SURVEY

In the recent decade, the application of Li-Fi technology has played a vital role in vehicular communication. Basically, the vehicles contain many lighting systems like headlights, taillights, and brake lights, which can function as Li-Fi transmitters [8-10]. Further, photodiodes can be mounted on these vehicles to receive modulated light signals. This arrangement facilitates Li-Fi technology apt for short-range vehicle-to-vehicle communication, where line-of-sight optical links can provide fast and reliable data exchange between the vehicles. By transmitting safety messages through visible light signals, Li-Fi helps different vehicles to convey various significant information like speed reduction alerts, distance between vehicles, braking signals, and road hazard warnings in real time [11-14].

In addition to communication, modern vehicle safety systems increasingly rely on sensor technologies to monitor the surrounding environment. Sensors such as ultrasonic modules and radar systems can detect nearby vehicles, measure inter-vehicle distance, and identify potential obstacles on the road [15-16]. When integrated with communication technologies, these sensing systems can significantly improve situational awareness and driver response. For instance, an ultrasonic sensor can measure the distance between vehicles and trigger automatic braking if the distance falls below a predefined safety threshold. Similarly, radar sensors can scan the surrounding environment and provide visual information about nearby objects or vehicles [17-18].

Motivated by these advancements, this research proposes a Li-Fi based vehicle-to-vehicle communication and safety system designed to improve road safety and driver awareness. The proposed system integrates Li-Fi communication with ultrasonic distance measurement, radar-based object detection, and an automatic braking mechanism to create a comprehensive vehicular safety framework. Li-Fi communication enables vehicles to exchange real-time safety messages, while onboard sensors continuously monitor the surrounding environment and detect potential collision risks. By combining communication and sensing technologies, the system aims to reduce rear-end collisions, enhance driver reaction time, and support the development of intelligent transportation systems for next generation connected vehicles.

III. PROTOTYPE DESIGN

Fig.1 demonstrates the architecture of a Li-Fi based V2V collision prevention system that integrates optical communication, radar sensing, and data fusion to develop road safety. The system consists of three major components: Vehicle A (transmitter), the communication intermediate, and Vehicle B (receiver). In the transmitting vehicle, multiple sensors such as radar, camera, and LiDAR constantly monitor the surrounding environment. The received signal is passed through a data fusion and V2V electronic control unit (ECU) to create consistent data. This data is communicated to nearby vehicles using a Li-Fi module, which converts digital data into optical signals, while a radar module grants further ranging information. At the receiving vehicle, a photodetector-based Li-Fi receiver gets the optical signal and sends it to the data fusion unit. The processed data is analyzed to determine whether a collision risk exists. If a threat is perceived, the collision prevention system triggers safety mechanisms like automated emergency braking assistance.

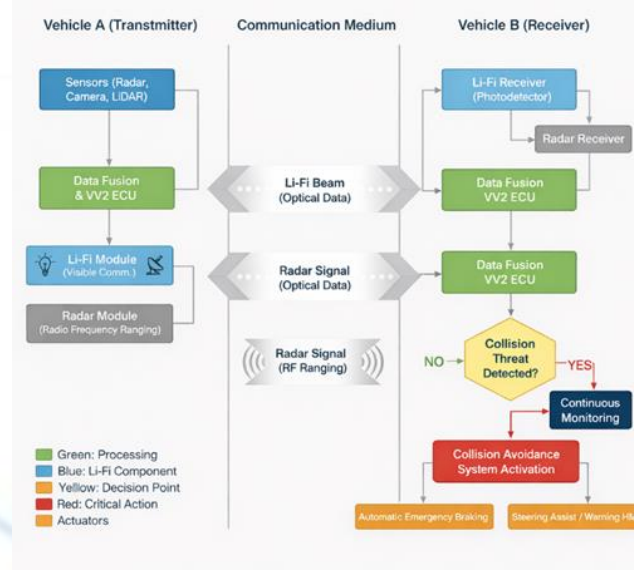


Fig. 1: Li-Fi enabled RADAR V2V collision prevention system

The fig. 2 shows the experimental model of the proposed Li-Fi supported V2V collision prevention system, realized using a small robotic vehicle platform. The prototype involves of a microcontroller-based control part incorporated with several sensing and communication units attached on a mobile framework. Ultrasonic sensors are arranged at the front of the vehicle to detect adjoining hurdles and evaluate the distance between the vehicles in real time. A Li-Fi transmission module, realized using light-emitting modules and optical receivers, facilitates optical data interchange between vehicles for mutual awareness. The administering unit coordinates sensor measurements, communication signals, and decision-making algorithms to determine possible collision risks. The structure also incorporates motor drivers and DC motors that manipulate the drive of the robotic vehicle. A display component is included to indicate system status and sensor readings through testing. This model setup is used to experimentally validate the feasibility of the suggested Li-Fi based V2V communication and collision avoidance mechanism under controlled laboratory conditions.

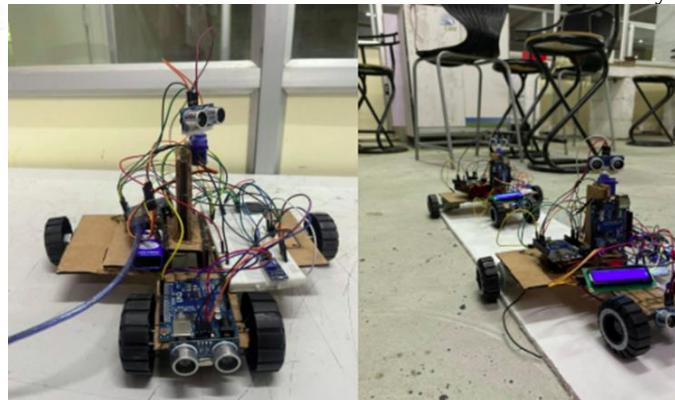


Fig. 2: Designed Model

IV. RESULTS AND DISCUSSIONS

The Li-Fi communication system was tested using a high intensity laser light module as the transmitter and an optical sensor unit as the receiver. Experimental observations show that the highly directional nature of laser-based Li-Fi enables stable and error-free communication between vehicles. As shown in fig. 3, the setup contains a Li-Fi transmitter-receiver module interfaced with a microcontroller and a 16×2 LCD display to track the real-time status of the vehicles. When no obstacle is identified, the model receives the signal and the LCD display shows “Clear Ahead,” indicating safe driving environment. On the other hand, when a potential hazard is detected by the transmitting vehicle, an alert signal is transmitted through the Li-Fi link. The receiving unit handles the signal and flashes a “STOP!” alert to alert the subsequent vehicles. The communication was found to be reliable even during daytime conditions, as the laser beam maintained sufficient intensity to overcome ambient light interference. The system successfully transmitted real-time warning messages such as speed reduction alerts and safety notifications with negligible delay. This low-latency communication significantly improves driver reaction time when compared to conventional RF-based communication systems.

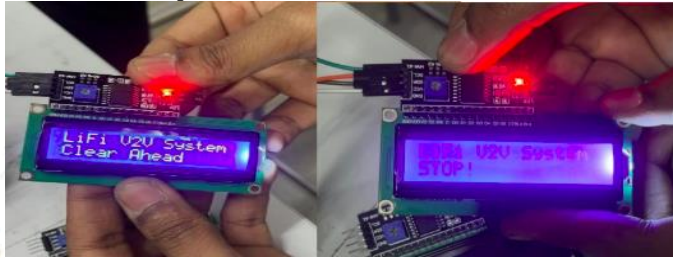


Fig. 3. The LCD module displaying stop and clear head indications.

Fig. 4 shows the radar-based exposure and visualization of the near-by vehicles implemented in the proposed collision prevention system. The setup contains a miniature radar sensing module interfaced with a microcontroller and an OLED display unit for real-time monitoring and visualization of surrounding vehicles. The radar sensor constantly scans the neighbouring environment and determines the relative distance and position of adjoining vehicles. In this figure, the detected vehicles are pointed as red markers on the radar display, revealing the existence and estimated location of vehicles within the sensing radius. The concentric circular rings denote different distance levels from the sensing unit, thereby allowing estimation of object proximity. The rotating green scanning line simulates the radar sweep that continuously monitors the surroundings. As the radar beam scans the environment, detected objects are updated on the display to provide situational awareness. This radar visualization unit aids the vehicle control system in recognizing nearby vehicles and helps in a timely decision-making for possible collision avoidance.

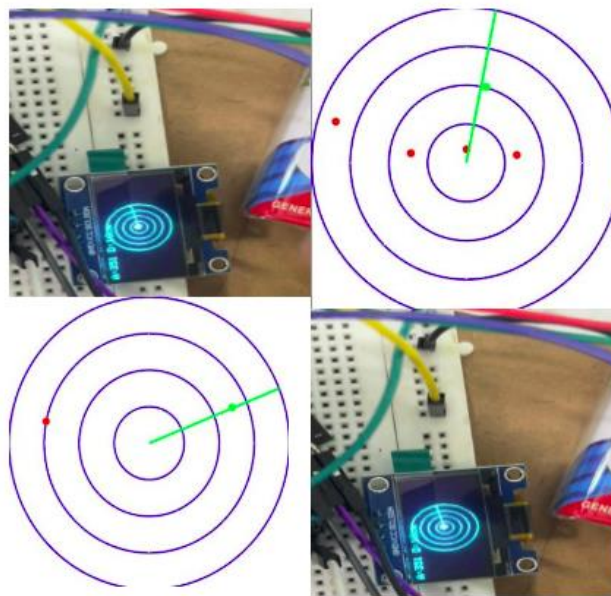


Fig. 4: Radar system displaying surrounding vehicles.

The fig. 5 explains the variation between the vehicle speed and inter-vehicle distance observed in the proposed collision avoidance system. The x-axis denotes the distance between vehicles, whereas the y-axis indicates the speed of the vehicle. When the inter-vehicle distance decreases below 20 cm, the system initiates a STOP command to instantly stop the vehicle, guaranteeing the safety. As the distance increases further than this threshold, the vehicle speed gradually increases. At approximately 35 cm, the system reaches the secure distance zone, where the vehicle is permitted to run at its highest speed of 60 cm/sec. This control logic facilitates in maintaining a safe spacing among the vehicles and decreases the probability of accidents.

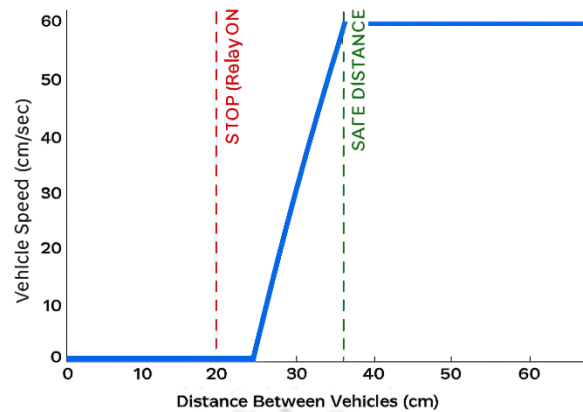


Fig. 5: Analysis of vehicle speed vs distance between the vehicles.

Table 1: Radar Detection performance

Target distance	Detection accuracy
20 cm	98%
40 m	96%
60 cm	94%

Table 2: Distance Measurement Accuracy

Actual distance	Measured distance	Error
20 cm	19.5 cm	0.5 cm
30 m	29.7 cm	0.3 cm

The table 1 illustrates the accuracy of the detection by the sensing module using the radar technology in relation to the distances from the target object. From the results, it is clear that the accuracy of the radar technology in sensing obstacles is high, especially in close-range sensing applications. For instance, at a distance of 20 cm, the accuracy of the system is 98%, indicating high accuracy in close-range sensing applications. However, the accuracy decreases slightly to 96% and 94% at distances of 40 cm and 60 cm, respectively, mainly due to signal attenuation and interference from environmental factors. The table 2 evaluates the accuracy of the ultrasonic distance measurement in the proposed system. For an actual distance of 20 cm, the sensor measured 19.5 cm with an error of 0.5 cm. At 30 cm, the measured value was 29.7 cm, resulting in an error of 0.3 cm, indicating reliable short-range distance sensing. Finally, we compared the vehicle localization accuracy achieved using three different approaches such as GPS-based localization, sensor-only localization, and the proposed Li-Fi based sensor fusion approach, which is depicted in fig. 6. The horizontal axis shows the localization techniques, while the vertical axis depicts the localization error in cm. The outcomes confirm that the GPS-only technique delivers the highest localization error due to signal degradation in urban environments. The sensor-only method enhances the accuracy by using onboard sensing devices. However, the proposed Li-Fi and sensor fusion method greatly decreases the localization error, indicating improved positioning reliability. This improvement emphasizes the efficiency of integrating Li-Fi communication with sensor fusion for accurate vehicle localization.

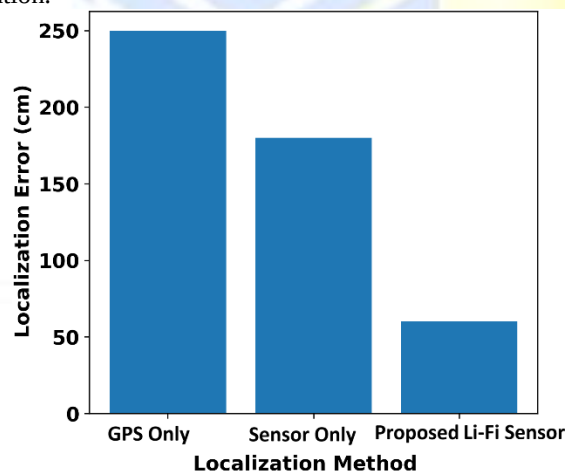


Fig. 6: Comparison of localization error with traditional methods

V. CONCLUSIONS

In conclusion, the present research demonstrates a Li-Fi technology-based vehicle-to-vehicle communication and collision avoidance system for enhancing the safety and accuracy of localization in smart transportation scenarios. In the proposed system, the Li-Fi communication technology, along with the radar sensing technology and ultrasonic distance measurement technology, is utilized for facilitating the real-time communication between the vehicles. Through the experimental implementation of the Li-Fi technology using the robotic vehicle, the effectiveness of the Li-Fi communication was proved for facilitating the communication between the vehicles. The radar visualization module improves the environmental awareness and enables the control unit to identify potential collision threats. Moreover, the results of the comparative localization analysis validate the effectiveness of the proposed Li-Fi assisted approach in reducing the localization error compared to the GPS-based and sensor-only approaches. The results clearly indicate that the proposed

approach for cooperative perception using optical communication can provide accurate positioning for vehicles, thereby improving the response time for hazardous scenarios.

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